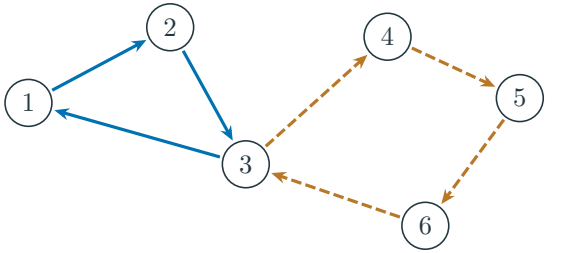
Crossing graph



C_1: $\sum_{e\in C_1} \alpha_e = \pi$ $\quad$ C_2: $\sum_{e\in C_2} \alpha_e = \pi$

group

Cycle inequality

$$Q_C(h) = \sum_{e\in C} \cot(\alpha_e)\,(h_q - h_p)^2 \;\geq\; 0$$

$$\sum_{e\in C} \alpha_e = \pi$$

equality $\iff$ h is constant on C

Assemble the Hessian

$$H_f = \sum_C H_C \succeq 0$$

$$\ker H_f = \operatorname{span}\{\mathbf{1}_C : C \text{ a component}\}$$

positive definite after gauge fixing

Edge-local inspection is misleading

$\cot(\delta_e)$ may be negative when $\delta_e > \pi/2$. Convexity is not an edge-by-edge statement.

Logical order

strand diagram / window induction $\longrightarrow$ winding-one partition $\longrightarrow$ cyclewise PSD $\longrightarrow$ strict convexity on $\Theta_f^{>0}$

The cycles are edge-disjoint; they may share vertices. Component shifts are the only common equality directions.